

MUNEEB AHMAD

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<https://mun-ahmd.github.io/apples-and-bananas>

Davis, California
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EDUCATION

University of California, Davis, CA, USA

Sept '25

Master of Science in Electrical and Computer Engineering

- June '27

Coursework: Modelling & Optimization in Computer Engineering, High Performance Statistical Computing, Computer Architecture, Information Theory & Coding

Advisor: John Owens

Jamia Millia Islamia, New Delhi, India

2021

Bachelor of Technology in Computer Engineering

- 2025

CGPA: 9.05 out of 10

Coursework: Computer Networks, Network Security, Computer Architecture, Digital System Design, Embedded Systems, Operating Systems, Parallel & Distributed Computing

SKILLS

- **Languages** - C, C++, CUDA, Triton, Python, Rust, GLSL, HLSL, JavaScript, SQL
- **Frameworks** - PyTorch, Tensorflow, vLLM, Vulkan, DirectX12, OpenGL, OpenMP, MPI, STM32-HAL
- **Tools** - NVIDIA Nsight, PyTorch profiler, RenderDoc, PIX, Visual Studio, perf, CMake, Perfetto

WORK EXPERIENCE

GPU Research Intern, Qualcomm, San Diego

Jun '26 -

Sept '26

- Analyzed ray-tracing performance on Adreno GPUs for both Vulkan and DirectX12 on both ray pipeline and ray query workloads.
 - Identified bottlenecks using hardware performance counters and traced fixes in the shader compiler codegen.
 - Fixes resulted in a 2x speedup in targeted configurations.
 - Identified memory overallocation bugs in compiler.
- Proposed memory layouts that achieve 100% (2x existing) memory bandwidth utilization for ray-tracing internal data by improving coalescing behavior.
 - Proposed design also improves cache-friendliness.
- Created AI agent-based tooling to perform end-to-end on-device performance analysis, regression testing, and root-causing.

Graduate Student Researcher at [Owensgroup](#), UC Davis

Nov '25 -

Present

- Created optimized custom GPU kernels for compressed paged attention with fused RoPE using Gluon and Triton.
- Integrated custom attention paths into vLLM, including managing the KV Cache in bespoke formats.
- Implemented gesvda on top of ROCsSolver for 10x faster approximate singular value decomposition on AMD GPUs.
- Discovered hardware efficient methods to compress KV Cache 8-30x with minimal accuracy degradation (<2% on task performance).
- Extensively evaluated LLM task performance using models like Llama and Qwen3 on benchmarks like Longbench, and gsm-8k.

Researcher at [BeyondDefence Lab](#), University of New Mexico

Jan '25

- Sept '25

- Creating a framework to find the optimal deployment strategy for a given deep learning model on a given edge device. Target use cases include satellites and robotics.
- Achieved a ~400% improvement in latency on the Coral EdgeTPU dev board.
- Worked with hardware accelerators such as Coral EdgeTPU and Nvidia Jetson Nano.
- Applied quantization to models and transformed formats for compatibility with hardware.

PROJECTS

SpMV on GPUs using DirectX Work Graphs [Github](#)

July '25
- Oct '25

- Implemented load balanced Sparse Matrix Vector Multiplication (SpMV)— an important operation used to accelerate deep learning models to the GPU — using DirectX12 Work Graphs to achieve better performance despite the irregular nature of the application.

Vulkan Forward Clustered Renderer [Github](#)

April '23
- July '24

- Implemented a forward-clustered renderer using Vulkan to handle large numbers of lights.
- Compute shader determines lights relevant to a cluster of space for optimal processing.
- Voxel renderer with compute shader based meshing to reduce draw count by 6 times.
- Focus on easy to modify rendering pipeline.

KV Cache Compression with KV Press [Github](#)

Sept '25 -
Dec '25

- Implemented KV cache eviction policy that considers recent attention scores and value vectors.
- Integrated with the kvpress framework
- Benchmarked using lm-eval-harness on Longbench using Mistral-7B-Instruct LLM.